

Circuit to Code

Hands-on electronics lessons toward programming and embedded projects.

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Source and updates: <https://github.com/LalatenduMohanty/circuit-to-code>

Source: beginner.md

Circuit to Code

Welcome to the hardware lab! Whether you already code in Scratch, build with LEGO Spike Prime, or are just getting started with electronics, this guide helps you learn circuits and components — the foundation for programming real embedded projects.

In coding, you build with blocks or text. Here, you are going to build with electrons. This guide has **Mission 0 (safety)** plus **8 circuit missions**. For each circuit, you will wire the parts, run an experiment, and observe what happens to discover the hidden rules of electronics.

You're ready for the full sequence — including transistors, logic, and an oscillator — so treat every circuit as a real mission, not a demo.

Blueprints in this guide use **real electrical schematic symbols**. Learn them once in Mission 0, then read every circuit like a pro.

What you need

Power (needed with either option)

Starter kits do **not** usually include a battery. Also get:

Part	Quantity	Notes
9V battery	1	Alkaline is fine
9V battery adapter	1	Snap connector to DC barrel jack (fits the kit's breadboard power module)

Plug the adapter into the power module's barrel jack, set the module to **5V**, then use the module's rails for **+5V** and **GND**. Unplug or switch the module off before changing wires.

Option A: Electronic Starter Kit (easiest)

Electronic Starter Kit — a collection of essential parts like resistors, LEDs, and jumper wires.

- [Amazon option 1](#) or [Amazon option 2](#)

These kits usually include a breadboard, 5V power module, jumper wires, LEDs, resistors, buttons, a potentiometer, a photoresistor, capacitors, and NPN transistors.

Kit notes for this guide

- Many kits ship **2N2222** transistors instead of **2N3904**. For these missions, 2N2222 is a fine substitute.
- Circuit 8 asks for **470Ω** LED resistors. If your kit has **330Ω** instead (common), use those — the flasher still works.

Option B: Buy parts individually

If you prefer to shop à la carte, get at least this set (covers Circuits 1–8 with a little spare), **plus** the 9V battery and adapter above:

Part	Quantity	Notes
Solderless breadboard (830 tie-points)	1	Half-size boards also work
5V breadboard power supply module	1	Barrel jack for the 9V adapter (USB optional)
Male-to-male jumper wires	1 pack	Assorted lengths help
LEDs (5 mm)	4+	Red is fine; mix colors if you like
220Ω resistors	4+	LED bodyguards for most circuits
470Ω resistors	2	Circuit 8 LED protectors (330Ω OK if needed)
10kΩ resistors	3+	Transistor base resistors
Push buttons (breadboard tactile)	2	4-pin style that fits a breadboard
10kΩ potentiometer	1	Panel or trimmer with 3 pins
Photoresistor (LDR)	1	Light sensor
2N3904 NPN transistors	2	Or 2N2222
100μF electrolytic capacitor	1	Polarized — watch + / –
10μF electrolytic capacitors	2	Polarized — watch + / –

Mission 0: The Maker Rules

Before you plug in a single wire, remember these rules:

1. **Power Off First:** Always unplug your battery module before changing wires.
2. **The Rule of the Long Leg:** LEDs and capacitors have a direction. Electricity only flows through them one way. The **long leg** always points toward the Positive (+) side. For capacitors with a stripe: **long leg = +, stripe / short leg = – (GND)**.
3. **Resistors are Bodyguards:** An LED connected directly to 5V power will POP and burn out instantly! Always protect your LED by putting a **220Ω** resistor in its path.
4. **Never connect + straight to –:** That makes a short circuit. No “spark tests.”
5. **Stuck? Ask before flipping forever:** If an LED will not light, check the troubleshooting list below. Flip the LED once if needed — then ask for help.

Quick vocabulary

Word	Meaning
Circuit	A complete loop so electricity can travel out and back
GND (–)	Ground — the return path for current
Resistor	Limits current (protects LEDs and transistor bases)
Series	Parts in one chain; same current, shared voltage
Parallel	Separate paths from the same power rails
Transistor	An electronic switch controlled by a small base signal
Capacitor	Stores and releases charge over time
Schematic	A drawing of a circuit using standard electrical symbols

Mission 0.5: Learn the Electrical Symbols

Engineers don't draw photos of every part — they draw **symbols**. Once you know these, you can read almost any beginner circuit diagram.

Electrical Symbols Used in This Guide

Power and Ground

 Wire

+5V Positive supply



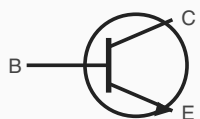
GND (Ground)

Tactile Push Button



Open until pressed

NPN Transistor (2N3904)



C (Collector)

E (Emitter to GND)

Resistor



220 ohm, 10k ohm

Potentiometer



Variable resistor (knob)

LED



Triangle to bar

Photoresistor (LDR)



Changes with light

Capacitor (polarized)



+ = long leg

Tip: LED current goes anode (long leg / triangle) to cathode (short leg / bar)

Schematics show connections. Your breadboard is where you plug them in.

How to read a schematic

1. Find **+5V** (top / start of the path) and **GND** (bottom / end of the path).
2. Follow the wires — a line means a connection.
3. Match each symbol to a real part on your breadboard.
4. Check direction for LEDs and polarized capacitors (arrow / + mark).

Symbol means...	On your breadboard...
Zigzag resistor	Colored bands resistor
LED triangle + bar	LED (long leg = anode / triangle side)
Switch gap	Tactile push button (4-pin kit button)
Pot with arrow	Knob (potentiometer)
Resistor in a circle + light arrows	Photoresistor (LDR)
Capacitor with +	Electrolytic capacitor
Circle with C/B/E	Transistor (2N3904)

Mission 0.6: Breadboard Basics

A breadboard is a plastic board full of holes that connect inside — like LEGO for wires.

- **Middle area:** Each short row of 5 holes (a–e, and f–j) is connected **side to side**. Parts in the same row touch electrically.
- **Power rails:** The long strips on the sides (often marked **+** and **–**) run the length of the board. Put **+5V** on the red/+ rail and **GND** on the blue/– rail.
- **Tip:** Leave a gap (the center ditch) between the left and right halves — they do **not** connect across that ditch.
- **Schematic ↔ breadboard:** The schematic shows *what connects to what*. The breadboard is *where* you plug those connections.

If it doesn't work (Troubleshoot)

Check these in order:

1. Is power plugged in and turned on?
2. Is there a complete loop from +5V back to GND?
3. Is the LED the right way around (long leg toward + / anode toward higher voltage)?
4. Is there a 220Ω resistor protecting the LED?

5. Are wires in the same breadboard row when they should touch?
6. For a tactile push button: does it **straddle the center ditch**, and are you using **opposite corners**? (Those pins connect only when pressed.)

Phase 1: The Basics of Flow

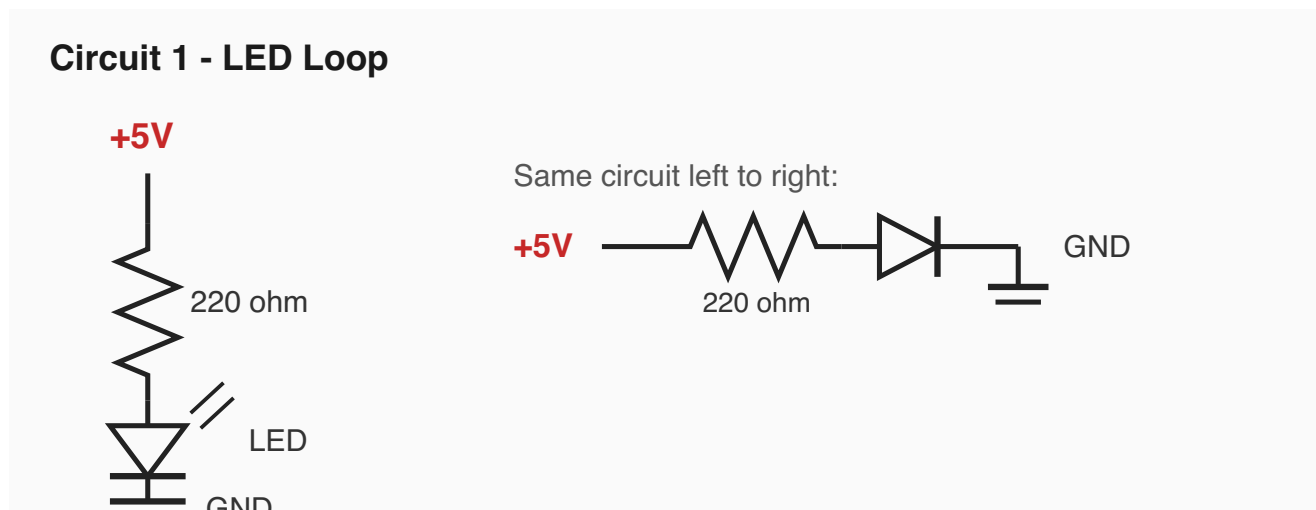
In Scratch, code runs in a loop. In electronics, electricity must complete a continuous circle (a circuit) from the Positive (+) power to the Negative (–) Ground.

Circuit 1: The Circle of Power (The Loop)

The Goal: Make an LED light up and learn how a circuit breaks.

Parts Needed: 5V Power, 1x 220Ω Resistor, 1x Red LED.

Schematic:



Symbols in this circuit: +5V · resistor · LED · GND

The Discovery (Observation)

Experiment: Once the light is on, pull one end of the resistor out of the breadboard.

What happened? The light turns off instantly.

The Concept: Electricity is like a train on a track. If there is a gap anywhere in the track, the train stops completely!

You built it! You learned: a circuit must be a closed loop.

Circuit 2: Press for Action (Tactile Push Button)

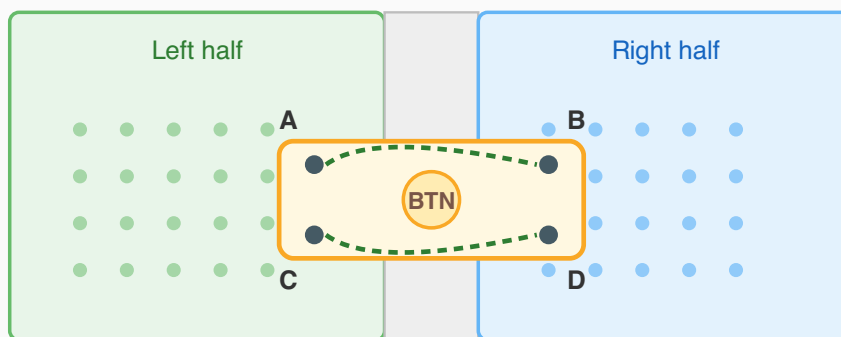
The Goal: Use the **4-pin tactile push button** from your starter kit to control an LED — the hardware version of `when space key pressed` in Scratch.

Parts Needed: 5V Power, 1x tactile push button (from the kit), 1x 220Ω Resistor, 1x Red LED.

Starter kits usually include several of these little square buttons with four bent legs. They click when pressed and spring back when you let go.

How to seat the kit button

Kit tactile button on a breadboard



Always connected

A to B, and C to D
(across the ditch)

Connect when pressed

Use opposite corners:
A to D, or B to C

1. Seat the button so it straddles the center ditch (two legs each side).
2. Wire opposite corners into your circuit - those pins join only while you press.
3. Pins across the ditch on the same row are already joined inside the button.

Do not use those as your switch - use opposite corners instead.

1. Push the button into the breadboard so it **straddles the center ditch** — two legs on the left half, two on the right.
2. Wire **opposite corners** into the circuit (for example top-left and bottom-right). Those pins connect **only while pressed**.
3. Pins across the ditch on the **same row** are already joined inside the button. That is normal — do not use those as your “switch.”

Schematic:

Circuit 2 - Tactile Button + LED



Symbols in this circuit: +5V · switch (tactile button) · resistor · LED · GND

The Discovery (Observation)

Experiment: Press the button. The LED lights up immediately. Let go, and it turns off.

What happened? Light on only while your finger is down — just like holding a key in Scratch.

Challenge: Move one wire to the *same row across the ditch* instead of an opposite corner. What happens? (The LED may stay on even when you are not pressing — those pins are already joined inside the button.)

The Concept: The tactile button is a mechanical drawbridge. When you press it, the bridge lowers, completing the “track” from Circuit 1 so the electrons can flow. When you let go, the spring opens the gap again.

You built it! You learned: a tactile switch can open or close a circuit.

Circuit 3: Chain vs. Path (Series and Parallel)

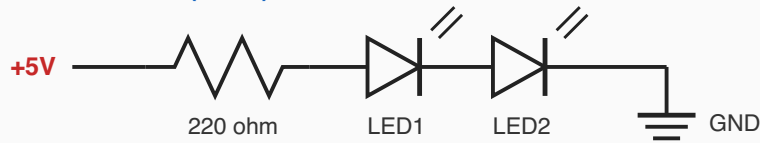
The Goal: See what happens when components have to share power.

Parts Needed: 5V Power, 2x LEDs, 2x 220Ω Resistors.

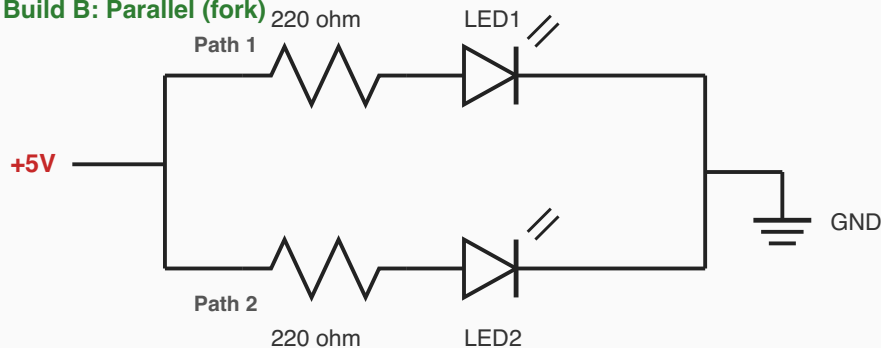
Schematic:

Circuit 3 - Series vs Parallel

Build A: Series (chain)



Build B: Parallel (fork)



Symbols in this circuit: two ways to connect the same parts — **series** (one path) vs **parallel** (forked paths)



The Discovery (Observation)

Experiment: Compare the brightness of the LEDs in Build A vs Build B.

What happened? In the Series chain, the lights are very dim (or may barely glow). In the Parallel fork, both are bright.

The Concept: In a Series chain, the LEDs have to share the 5V from the battery (roughly half each). In a Parallel fork, they each get their own path to the full 5V.

Challenge: What happens if you try 3 LEDs in series?

You built it! You learned: series shares; parallel keeps each path strong.

Phase 2: Sensors and Adjustments

Circuit 4: The Dimmer Knob (Potentiometer)

The Goal: Control the flow of electricity smoothly, like a slider variable in Scratch.

Parts Needed: 5V Power, 1x 10kΩ Potentiometer (knob), 1x 220Ω resistor, 1x LED.

Potentiometer tip: A pot has **3 pins**. Looking at the pins from the front (knob facing you), they are usually **Left — Middle (wiper) — Right**. For this dimmer, we use the pot as a **variable resistor** (rheostat): wire **one outer pin** and the **middle pin** into the circuit. Connect the unused outer pin to the middle pin with a short jumper (recommended), or leave it empty.

Schematic:

Circuit 4 - Potentiometer Dimmer (rheostat)



Symbols in this circuit: potentiometer (zigzag + arrow) · resistor · LED

(If turning the knob the “wrong way” makes it brighter when you expected dimmer, move the +5V wire to the other outer pin instead.)

The Discovery (Observation)

Experiment: Slowly turn the potentiometer knob back and forth.

What happened? The LED smoothly transitions from very bright to completely off.

The Concept: A potentiometer is a variable resistor. Turning the knob is like **pinching a hose** — the more you pinch, the less water (electricity) gets through, so the light gets dimmer!

You built it! You learned: resistance can change, and that changes brightness.

Circuit 5: The Sunrise Sensor (Photoresistor)

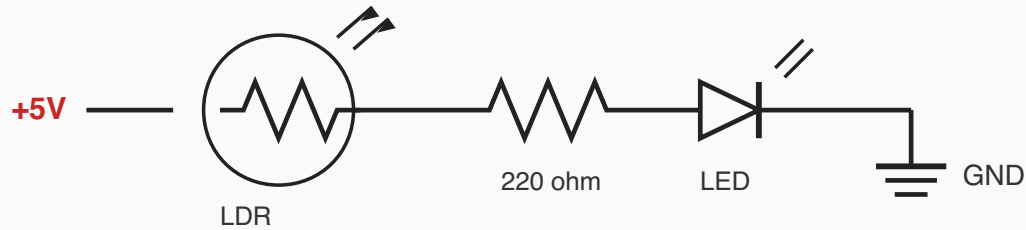
The Goal: Make a circuit that reacts to its environment.

LEGO Analogy: This is the same kind of idea as the light sensor in your Spike Prime kit!

Parts Needed: 5V Power, 1x Photoresistor (LDR), 1x 220Ω resistor, 1x LED.

Schematic:

Circuit 5 - Photoresistor (LDR) Sensor



Symbols in this circuit: LDR (resistor in a circle with light arrows) · resistor · LED

The Discovery (Observation)

Experiment: Cover the photoresistor tightly with your thumb to block the room light.

What happened? The LED gets dimmer or turns off when it gets dark.

The Concept: The photoresistor changes its resistance based on light. More light = lower resistance (brighter LED). No light = huge resistance (dim/off LED).

You built it! You learned: a sensor can change a circuit without a button.

Phase 3: Hardware Magic (Logic & Timers)

Circuit 6: Hardware Logic (The AND Gate)

The Goal: Build the Scratch idea of `if <A> and <B>` using physical hardware.

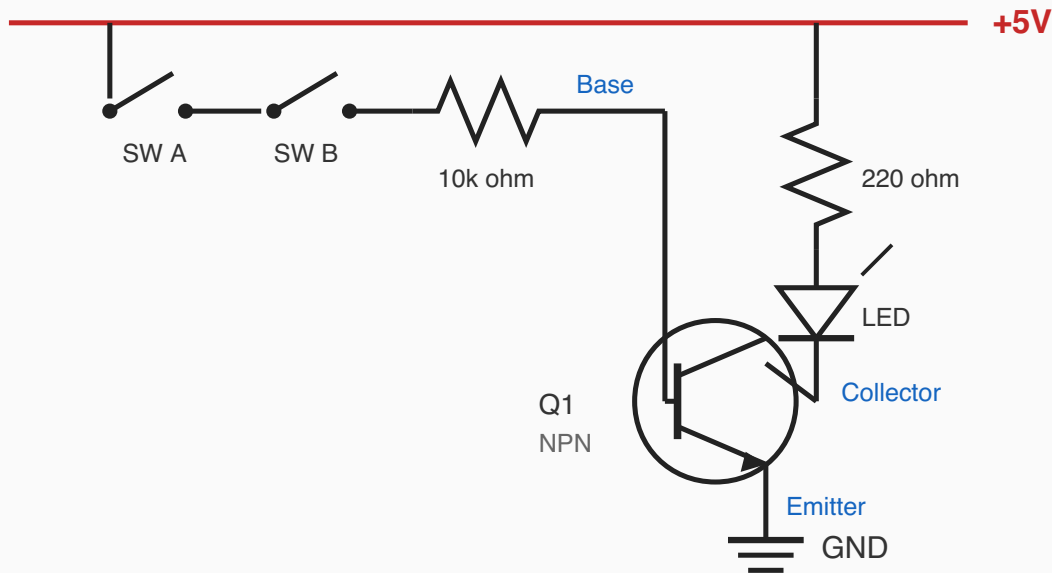
Pro tip: Power off, wire carefully, then power on. The 10kΩ base resistor protects the transistor — don't skip it.

Parts Needed: 5V Power, 1x 2N3904 NPN Transistor, 2x Push Buttons, 1x 10kΩ Resistor (base), 1x 220Ω Resistor (LED), 1x LED.

Transistor tip: Hold the flat face toward you. Pins left → right are usually **Emitter, Base, Collector** (check your transistor's datasheet/label if unsure). Emitter goes to GND.

Schematic:

Circuit 6 - Transistor AND Gate



Symbols in this circuit: two switches · 10kΩ · NPN transistor · 220Ω · LED



The Discovery (Observation)

Experiment: Press only Switch A. Then press only Switch B. Finally, press both switches at the same time.

What happened? The LED only lights up when BOTH are pressed.

The Concept: The transistor is an electronic switch. A small signal at the middle pin (the Base) lets a bigger current flow from Collector to Emitter. Both buttons must be pressed to deliver that Base signal — just like Scratch `and`.

You built it! You learned: hardware can do logic without writing code.

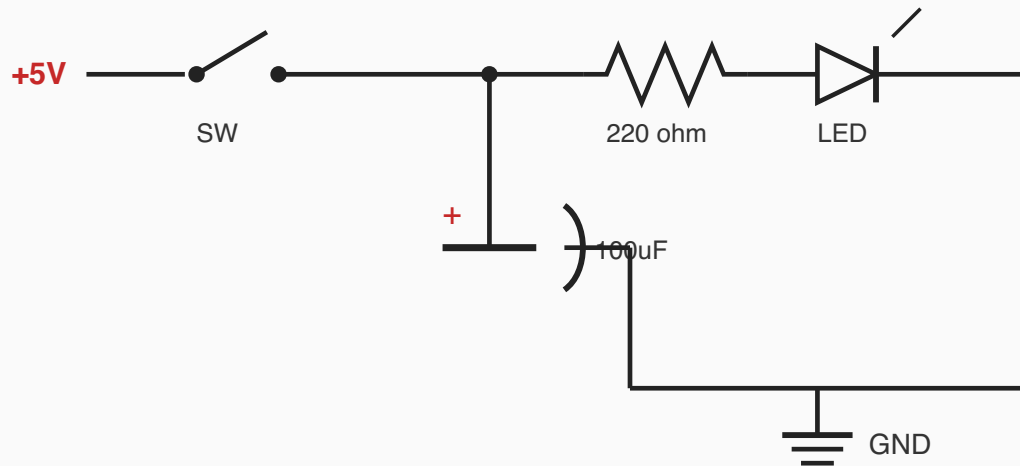
Circuit 7: The Battery Tank (Capacitor)

The Goal: Store electricity to keep the light on even after the switch is released.

Parts Needed: 5V Power, 1x Push Button, 1x 100μF Capacitor, 1x 220Ω Resistor, 1x LED.

Schematic:

Circuit 7 - Capacitor Battery Tank



Polarized capacitor: + (long leg) toward switch junction; stripe/short leg toward GND.

Symbols in this circuit: switch · polarized capacitor · resistor · LED

After you finish, unplug power. Capacitors can hold a little charge for a moment — that's why the LED fades!



The Discovery (Observation)

Experiment: Hold the button down for 3 seconds. Then let go.

What happened? Instead of turning off instantly, the LED fades out slowly!

The Concept: A capacitor is like a water tower. While the button is pressed, it fills up with electricity. When you let go of the button, the capacitor empties its stored electricity into the LED, keeping it alive for a few extra seconds.

You built it! You learned: capacitors store and release energy.

Circuit 8: The Flasher (Oscillators)

The Goal: Create an automatic timer loop without writing any code!

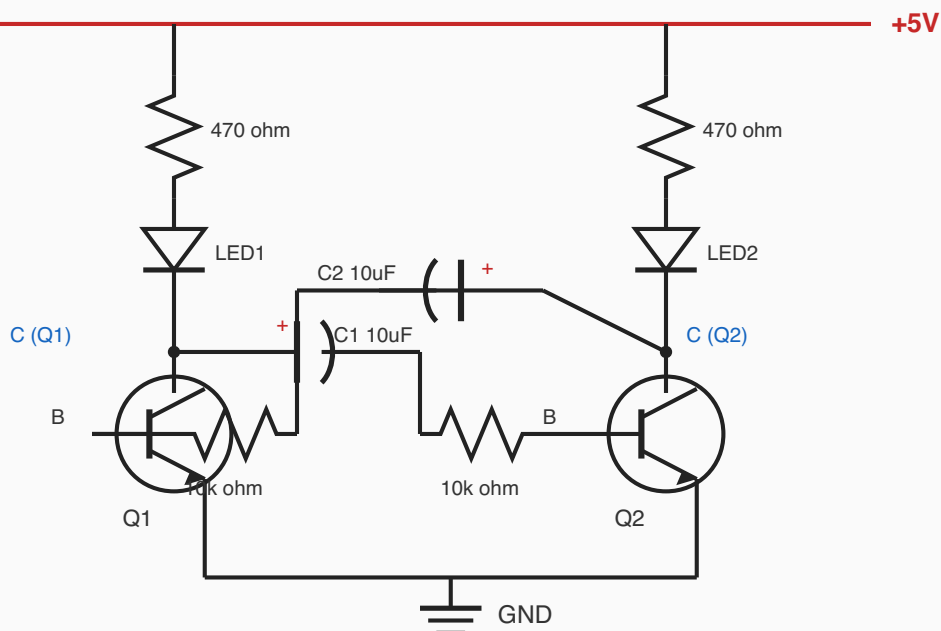
This is the most careful breadboard build in the guide — work slowly, follow the checklist, and double-check polarity on both capacitors.

Parts Needed:

- 5V Power
- 2x 2N3904 NPN Transistors (Q1, Q2)
- 2x $10\mu\text{F}$ Capacitors (C1, C2) — watch polarity: + toward the transistor collectors
- 2x LEDs
- 2x 470Ω Resistors (R_LED — LED protectors)
- 2x $10\text{k}\Omega$ Resistors (R_B — base resistors)

Schematic:

Circuit 8 - Astable Multivibrator (Flasher)



Symbols in this circuit: two NPN transistors · two polarized capacitors · four resistors · two LEDs

Build checklist (same as the schematic):

1. Both emitters → GND.
2. Each collector → its own LED → 470Ω → +5V.
3. C1 from Q1 collector toward Q2 base (through $10\text{k}\Omega$ at the base).
4. C2 from Q2 collector toward Q1 base (through $10\text{k}\Omega$ at the base).
5. Capacitor + toward the collector side; stripe toward the base side.



The Discovery (Observation)

Experiment: Wire this up carefully. You do not need to press any buttons. Just connect power and watch.

What happened? The LEDs flash back and forth automatically!

The Concept: The two capacitors take turns filling up and dumping out. When one dumps, it turns off its transistor and turns on the other. This creates a repeating pulse — a simple version of how electronic circuits can make a steady “tick-tock” rhythm (real computers use more precise clocks, but the idea of a repeating pulse starts here!).

You built it! You learned: circuits can create loops in hardware, not only in code.